

SUMMARY

Dataset Overview

After dropping unwanted column, The data had 545 housing records and 12 features such as area, bedrooms, bathrooms, stories, etc., including information on amenities like airconditioning, basement, and furnishing status. The data was clean and there was no missing value or duplicate row in the data

Data Preprocessing

One-hot encoding of categorical yes/no variables was done on these columns and were converted to numeric type. To provide the model with more inputs, features were engineered by incorporating other features.

Key findings from Exploration.

Key findings that the area is the most important factor in price, which was also indicated by the feature importance scores from the Random Forest model.

Model Building & Performance

Linear Regression was better than Random Forest, indicating that the relationship between features and price in this data set is more linear, in that a model using just a straight line would have been more successful than a complex ensemble.

Most Influential Features

The feature importance analysis revealed that area and bathrooms are the most significant factors affecting house prices, with the former having a greater impact than the latter.

Business Recommendation

In real estate investing, it is more prudent to invest in larger properties as opposed to upgrading the property with luxury amenities such as extra guestrooms or hot water heaters. Prioritize area over amenities